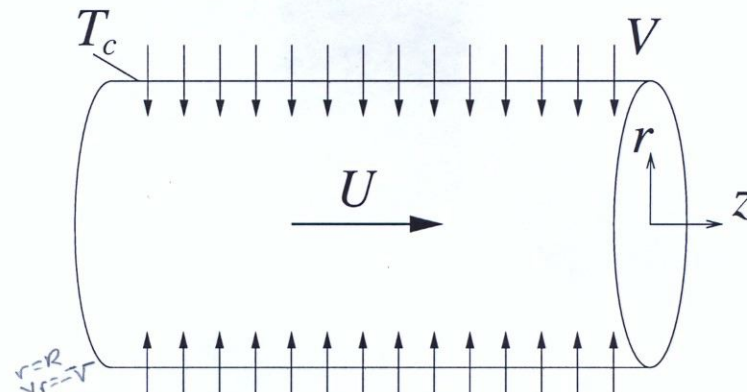


NS8 An infinitely long porous cylinder of radius R is surrounded by a liquid of density ρ , viscosity μ , specific heat c , and thermal conductivity k . The cylinder moves axially with velocity $v_z = U$ and entrains the liquid through its surface pores with a suction velocity $v_r = -V$, inducing an axisymmetric velocity field with components $v_r(r)$, $v_z(r)$ and $v_\theta = 0$. At large radial distances $r \gg R$ from the cylinder the liquid is at rest with uniform pressure p_∞ . Assume in the following analysis that the effect of gravity on the fluid motion can be neglected.

1. Use the continuity equation to determine the radial velocity component $v_r(r)$ for $R \leq r < \infty$.
2. From the radial component of the momentum equation compute the pressure distribution, demonstrating in particular that $\partial p / \partial z = 0$.
3. Determine the axial velocity distribution $v_z(r)$ as a function of the suction Reynolds number $Re = \rho V R / \mu$ (to solve the problem, assume that the two independent solutions of the resulting linear ODE are of the form $v_z \propto r^\alpha$, obtaining the two values of α that are compatible with the equation).
4. Obtain the force acting on the cylinder (per unit length).
5. Calculate the trajectory of the fluid particle located initially at $r = r_0$ and $z = 0$.
6. Compute the temperature field $T(r)$ when the cylinder temperature T_c is different from that of the surrounding atmosphere $T_\infty > T_c$. Neglect viscous dissipation and express the result with use made of the Prandtl number $Pr = \mu c / k$.
7. Determine the heat transferred to the cylinder per unit time (per unit length).

$$P_\infty \quad T_\infty$$



$$1) \nabla \cdot \vec{v} = 0 \Rightarrow \frac{\partial}{\partial r}(r v_r) = 0 \Rightarrow v_r = -\frac{R V}{r} \quad 2) \int v_r \frac{\partial v_r}{\partial r} = -\frac{\partial p}{\partial r} \Rightarrow P = P_\infty - \frac{\rho R^2 V^2}{2 r^2} \quad \frac{\partial p}{\partial z} = 0$$

$$3) z\text{-mom: } \int v_r \frac{d v_z}{d r} = \mu \frac{1}{r} \frac{d}{d r} \left(r \frac{d v_z}{d r} \right) \Rightarrow \frac{d}{d r} \left(r \frac{d v_z}{d r} + R e v_z \right) = 0 \quad r \frac{d v_z}{d r} = -R e v_z$$

$$4) \vec{F} = \int_{\Sigma_c} (\vec{e}_1 \cdot \vec{e}_r) d\sigma = -\mu U \int_{\Sigma_c} \frac{v_r}{r} z \pi R \cdot \vec{e}_z = -\frac{\rho U V z \pi R}{2} \vec{e}_z \quad v_z = C r^{-R e} = U \left(\frac{r}{R} \right)^{-R e}$$

$$5) \frac{d r}{d t} = v_r = -\frac{R V}{r} \Rightarrow r^2 - r_0^2 = -2 R V t$$

$$\frac{d z}{d t} = v_z$$

$$\frac{d r}{d z} = \frac{v_r}{v_z} \Rightarrow \frac{z}{R} = \frac{U}{(z - R e)^{1/R e}} \left[\left(\frac{r_0}{R} \right)^{2/R e} - \left(\frac{r}{R} \right)^{2/R e} \right]$$

$$6) \int v_r \frac{d T}{d r} = \frac{\mu U R}{r} + \frac{k}{r} \frac{d}{d r} (r \frac{d T}{d r}) \quad \begin{cases} T(R) = T_c \\ T(\infty) = T_\infty \end{cases} \quad \frac{T - T_\infty}{T_c - T_\infty} = \left(\frac{r}{R} \right)^{-Pr Re}$$

$$7) k \frac{dT}{dr} \Big|_{r=R} \cdot 2\pi R = 2\pi R \rho c V R (T_\infty - T_c)$$